Q1.

(a) The diagram shows a human sperm cell.



(i) What is the function of the nucleus of a sperm?

.....

.....

1 mark

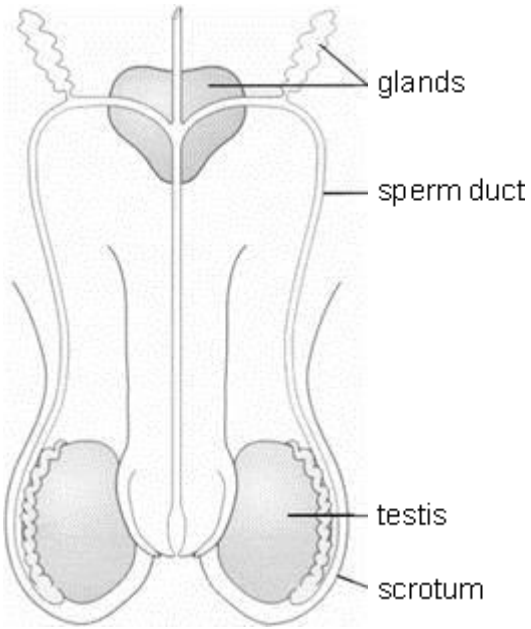
(ii) Sperm use their tails to swim to the egg so that fertilisation can take place. Give **one** other way in which a sperm is adapted for fertilisation. Explain your answer.

.....

.....

1 mark

(b) The diagram shows the human male reproductive system.



The testes are outside the main part of the body.
Suggest the relationship between temperature and sperm production.

.....

.....

1 mark

(c) A chemical called a hormone, which changes a boy's body, is produced from adolescence onwards.

(i) Where is this hormone produced?

.....

1 mark

(ii) Describe **one** change caused by this hormone.

.....

.....

1 mark

(d) (i) During sexual intercourse about 250 million sperm are released at a time into the body of a woman.
Why is it necessary to release so many sperm at a time?

.....

.....

1 mark

(ii) Before being released from the body, sperm are mixed with a fluid. The fluid contains sugar.
Explain how **and** why a sperm makes use of the sugar.

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.....

.....

2 marks
Maximum 8 marks

Q1.

(a) (i) any **one** from

- it controls the activities of the sperm **or** cell
- it carries **or** passes on genetic information
  accept 'passes on information'
- it contains the genes
  accept 'it contains chromosomes'
  *do **not** accept 'it carries all the genetic information of the baby'*
- fertilisation

1 (L6)

(ii) **Both the adaptation and the reason are required for the mark.**

any **one** from

- streamlined **or** has a pointed head to help it swim **or** penetrate the egg
- it produces a chemical to help it penetrate the egg
- very little cytoplasm so it swims faster

1 (L7)

(b) the best temperature for making sperm is lower than body temperature
accept 'a lower temperature is needed for producing sperm'
accept 'at higher temperatures sperm might be defective'

1 (L7)

(c) (i) in the testes **or** testis

1 (L7)

(ii) any **one** from

- body hair **or** pubic hair **or** hairy armpits
  *do **not** accept 'acne **or** spots'*
- facial hair **or** beard
- breaking **or** lowering of the voice
- growth of the penis or testes
  do not accept 'the testes drop'
- body becomes more muscular
  *accept 'shoulders broaden' **or** 'muscles develop'*
- sperm production begins

- growth spurt
  *do **not** accept 'grows taller'*

1 (L6)

(d) (i) any **one** from

- to increase the chance of one reaching the egg **or** the oviduct
  accept 'to make sure an egg is fertilised'
- most sperm die before reaching the egg **or** oviduct
  accept 'most do not reach the egg'
  'most sperm die' is insufficient

1 (L7)

(ii) for respiration
accept 'for energy'
*do **not** accept 'for food'*

1 (L7)

for movement **or** for swimming
accept 'to reach the egg'

1 (L7)

[8]